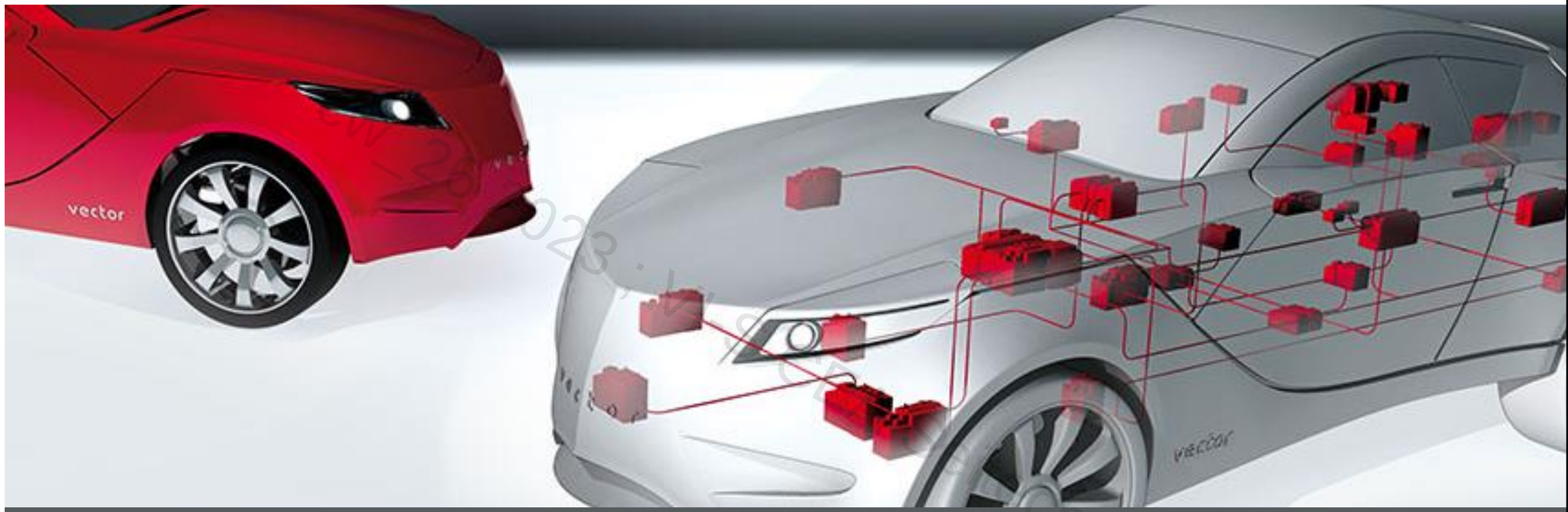
VECTOR



AUTOSAR in Practice

Diagnostics

Agenda

- **Diagnostics**
  - CANTP – CAN Transport Layer
  - DCM – Diagnostic Communication Manager
  - DEM – Diagnostics Event Manager
  - FIM – Function Inhibition Manager
  - Coming up next

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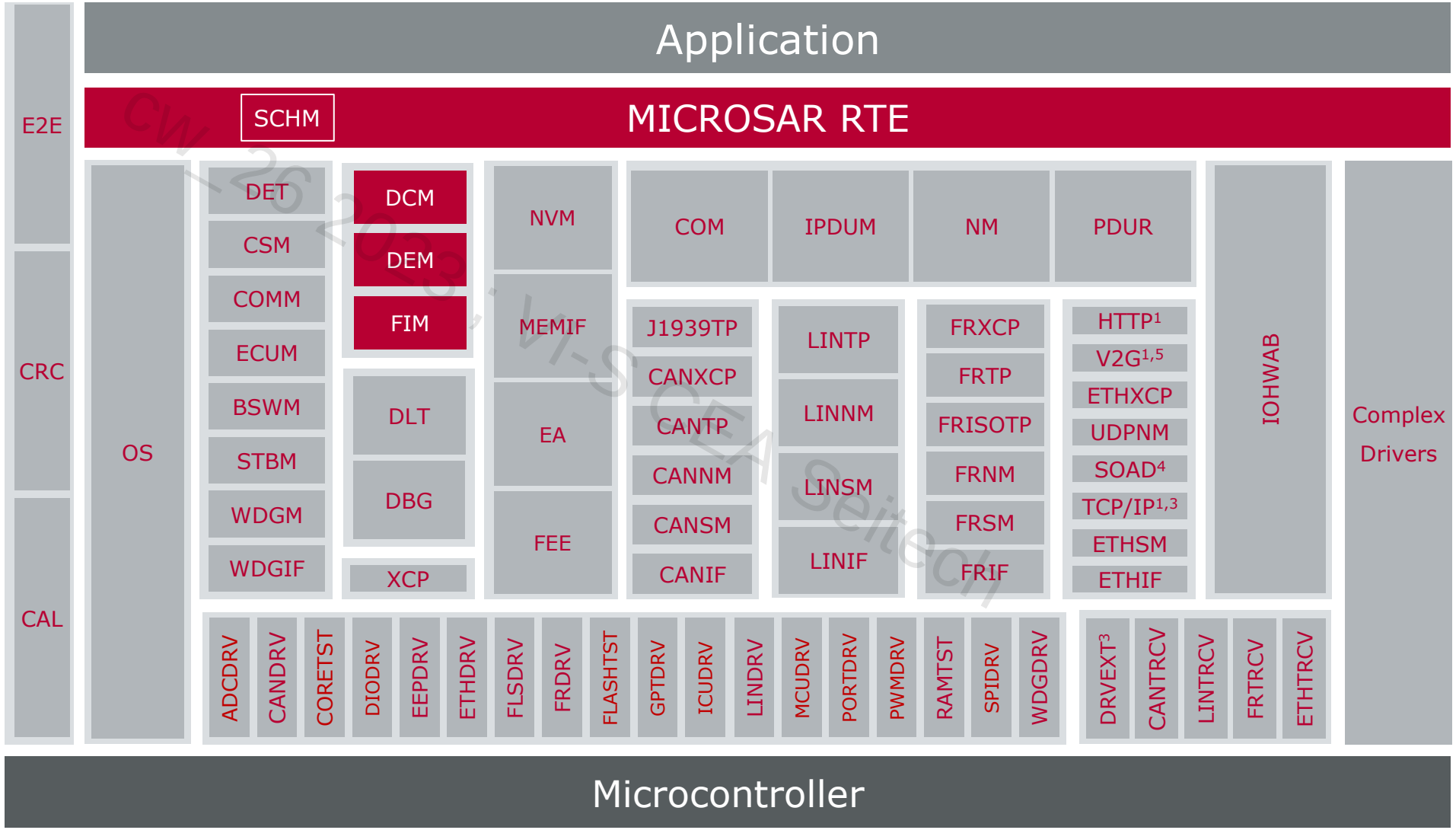
Diagnostics provided by

- DCM
- DEM
- FIM

Required modules

- CANTP
- FRTP
- LINTP
- SOAD (DoIP)



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ISO 14229-1

- Also known as: UDS = Unified Diagnostic Services
- Definition of diagnostic protocol with services
- Network-independent
- Definitions
  - A “diagnostic protocol” is necessary for communication between a diagnostic tester and an ECU
  - A diagnostic protocol incorporates a set of diagnostic services that are transmitted serially between the diagnostic tester and ECU(s)
  - A diagnostic protocol contains data and information on the meanings of the messages to be sent and the ECU’s behavior in response to a message

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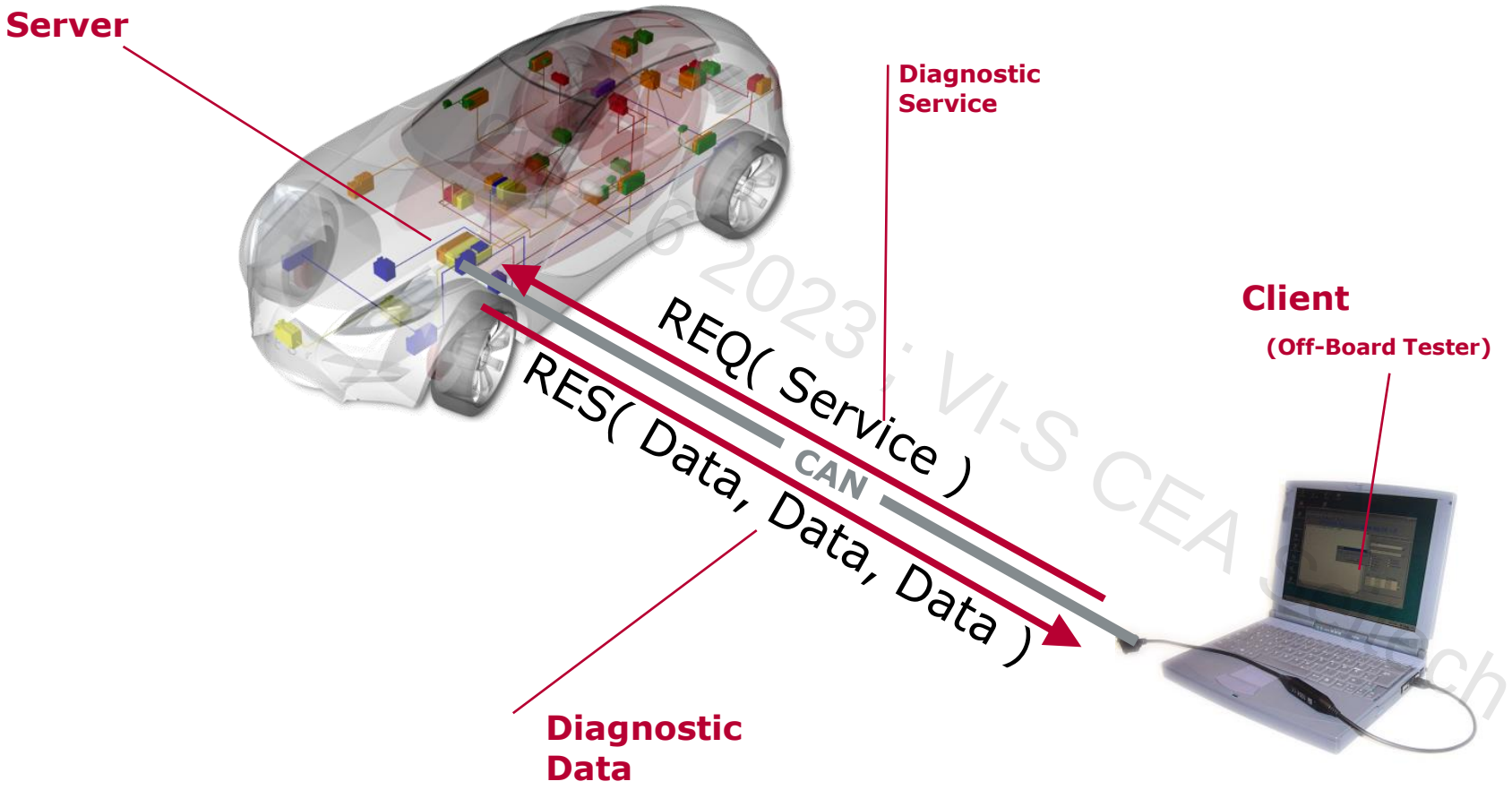
Network View of Diagnostics



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Network View of Diagnostics

A **Diagnostic Request** is either **functional** or **physical**. This terms refer to the addressing strategy. Functional addressing is a "broadcast" for all ECUs; physical addressing is peer-to-peer communication (point-to-point).

- A **physical request** will be answered **physically** as this accounts for the peer-to-peer property of the addressing.
- A **functional request** will also be answered **physically**, since the tester has to differentiate the responses of all ECUs from each other.

Diagnostic Requests (REQ) and **Diagnostic Responses (RES)** may already contain data, but this is not necessarily the case and depends on the definition of the diagnostic service.

A diagnostic request will typically be followed by a diagnostic response. If there is no response at all the tester will observe a timeout. Some services, like the functional service Tester Present, do not need any response from the ECU at all.

The **Diagnostic Response (RES)** can be one of the following:

- **Positive Response:** The service request has been processed correctly and a response, possibly with data, is on its way to the tester.
- **Negative Response with Negative Response Code (NRC):** The service processing encountered an error specified by the NRC code, and this is indicated to the tester.
- **Negative Response with NRC RequestCorrectlyReceiveResponsePending (RCRRP):** The diagnostic server needs more time to process the request and can not yet deliver a positive or negative response. According to UDS a diagnostic server can respond with RCRRP indefinitely. However, all services are expected to be processed within a finite time. The protocol is not finished unless the ECU answers with a final positive or negative response.

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DID – Data Identifier

	Services	Sub-Service (Identifier)
ReadDataByIdentifier	$22	F190
WriteDataByIdentifier	$2E	F190

read/write
e.g. VIN

Classical view (communication protocol oriented)

e.g. **VIN** Vehicle Identification Number

≙

DID — Data Identifier (e.g. Did 0x10)

AUTOSAR view (object oriented)

- ReadData
- WriteData
- ReadDataLength
- ConditionCheckRead
- ReturnControlToECU
- ResetToDefault
- FreezeCurrentState
- ShortTermAdjustment
- GetScalingInformation

Every DID has a Superset of Possible Operations

Normally only a subset is available for a certain DID

In AUTOSAR configuration only DIDs are available.

A DID is "translated" into UDS. Every Operation on a specific DID is translated into a Service and Service Identifier, i.e., the appearance on the bus is the same as non-AUTOSAR UDS.

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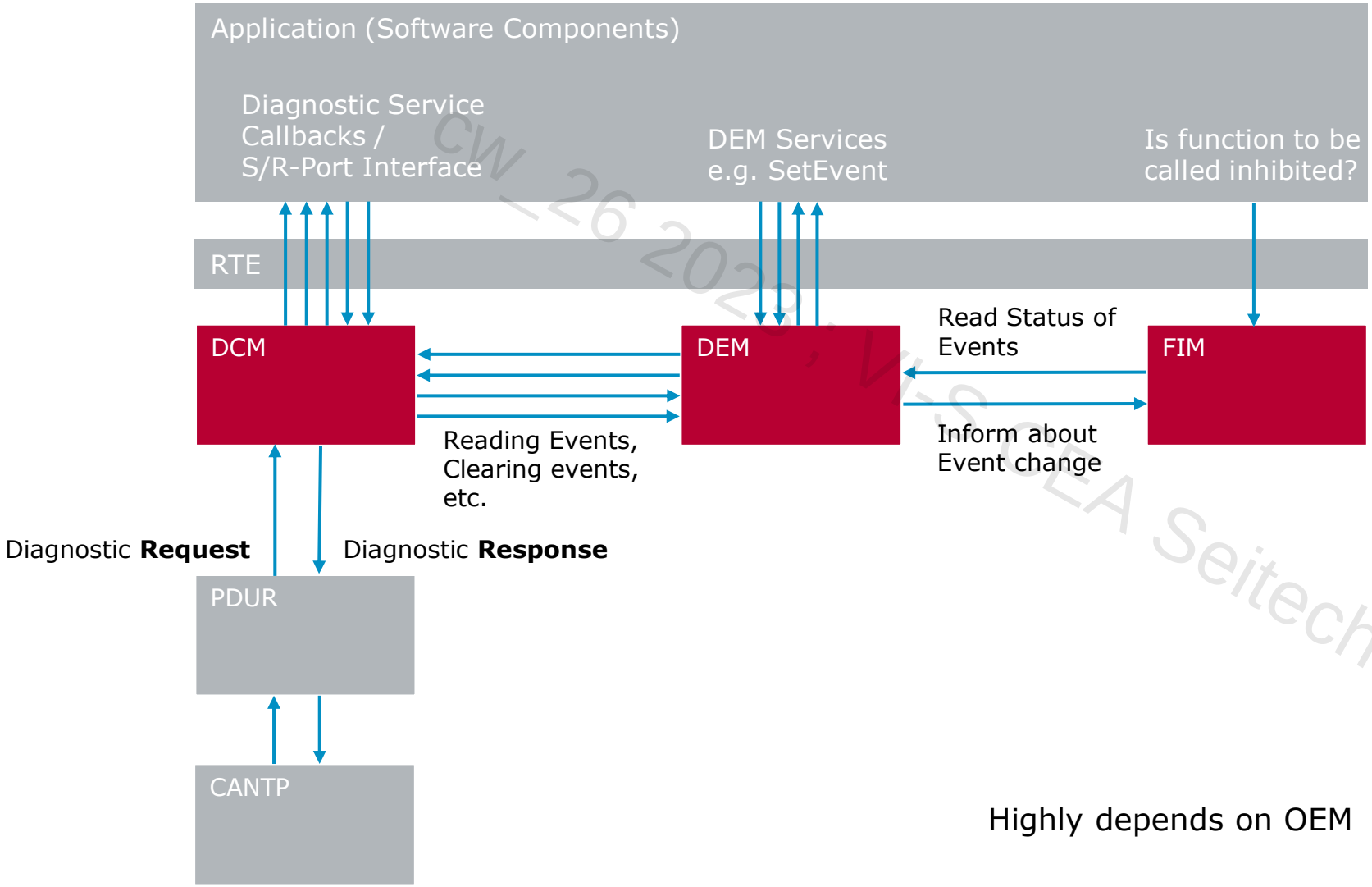
DCM – DEM - FIM



Highly depends on OEM

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Agenda

Diagnostics

- **CANTP – CAN Transport Layer**

DCM – Diagnostic Communication Manager

DEM – Diagnostics Event Manager

FIM – Function Inhibition Manager

Coming up next

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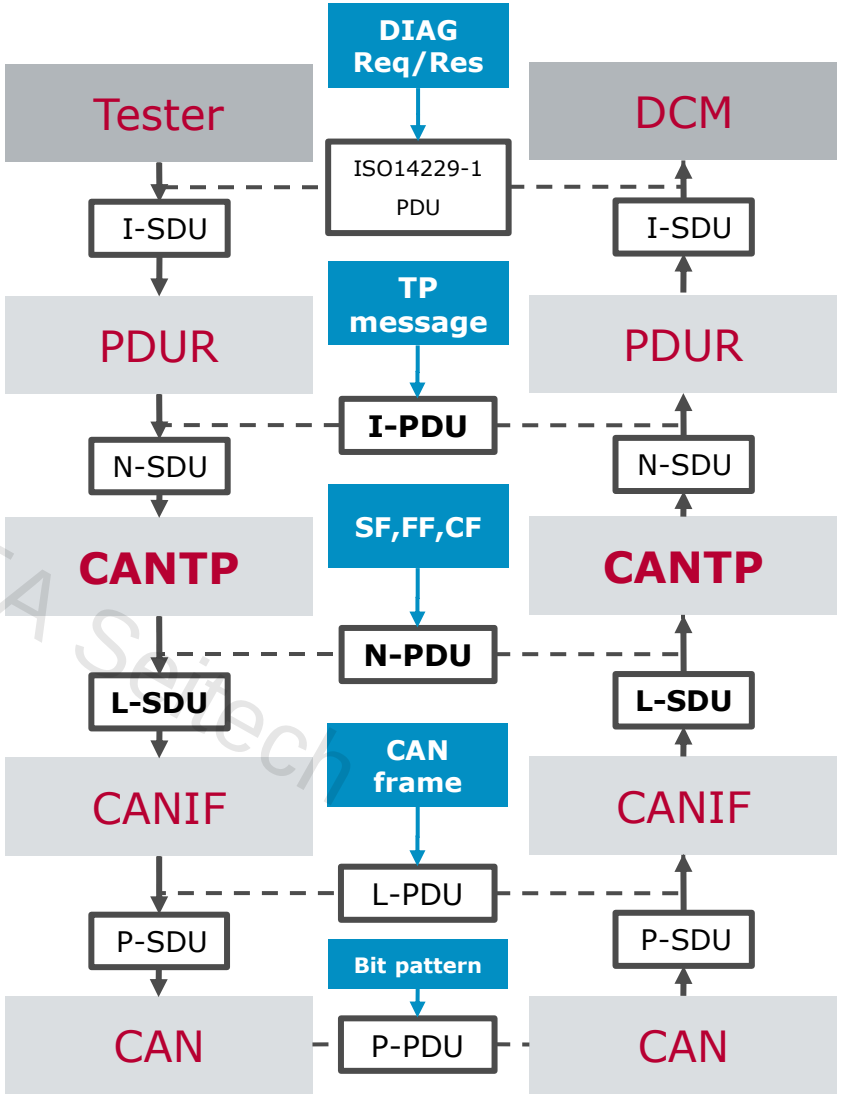
Module short information

- CANTP is the module between the PDU Router and the CAN Interface module
- The main purpose of the CAN TP module is to segment and reassemble diagnostic I-PDUs longer than 8 bytes
  - A TP message is called an N-SDU (Network-layer Service Data Unit)
  - Handling of Single (SF), First (FF) and Consecutive Frames (CF) uses N-PDUs (L-SDUs)
- Support of different addressing formats
  - Functional and physical
  - Normal, extended, and mixed



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Module short information

N_SA	Network Source Address
N_TA	Network Target Address
N_TAtype	Network Target Address Type
Mtype	Message Type diagnostics, remote diagnostics
N_PCI	Network Layer Protocol Information
N_Data	payload

Normal addressing format

A unique CAN identifier is assigned to each combination of N_SA, N_TA, N_TAtype and Mtype. N_PCI and N_Data are filed in the CAN frame data field.

Extended addressing format

A unique CAN identifier is assigned to each combination of N_SA and Mtype. A unique address is filed to each combination of N_TA and N_TAtype in the first data byte of the CAN frame data field. N_PCI and N_Data are filed in the remaining bytes of the CAN frame data field.

Mixed addressing format

A unique CAN identifier is assigned to each combination of N_SA, N_TA, N_TAtype. N_AE is placed in the first data byte of the CAN frame data field. N_PCI and N_Data are placed in the remaining bytes of the CAN frame data field.

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Agenda

- Diagnostics
- CANTP – CAN Transport Layer
- ▶ **DCM – Diagnostic Communication Manager**
- DEM – Diagnostics Event Manager
- FIM – Function Inhibition Manager
- Coming up next

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Overview

- DCM is network independent
- Provides a common API for diagnostic services
- The functionality of this AUTOSAR-Basic-SW is used by external diagnostic tools e.g. in the development, manufacturing, service or legislative OBD
- The DCM handles different (application layer) **diagnostic protocols**
  - OBD (ISO 15031-5)
  - Enhanced diagnostics (ISO 14229-1)
- For diagnostic **session handling** the network independent sections of the following specifications are also handled by the DCM
  - ISO 15765-3: Implementation of unified diagnostic services (UDS on CAN)
  - ISO 15765-4: Requirements for emissions-related systems

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Overview

OBD (On Board Diagnostics) refers to legislative diagnostics requirements defined by the respective country in which the vehicle is on the road. OBD is related to the so called "emission related" diagnostics.

In OBD, the diagnostic tester is often referred to as the "generic scan tool."

Among other things, OBD requires:

-Parallel operation to the "ordinary" UDS diagnostic communication

-Special UDS services with their own semantics defined for OBD diagnostics only. Examples:

-DCM: OBD communication services $01 - $0A (defined in SAE J1979 Rev May 2007)

-DEM: a special OBD FreezeFrame "record 0"…

-Implementing the above requirements in DEM and DCM, AUTOSAR OBD functionality meets all **light duty OBD regulations** in the world

-EOBD, Japan OBD, California OBDII, …

-Open issues, not treated in AUTOSAR

-MIL (Malfunction Indicator Lamp): blinking, light bulb check of the MIL, etc.

-Misfire fault handling: Debouncing and filtering

-No description on how to achieve OBD compliant diagnostic applications concerning state handling or diagnostic algorithms.

-This is due to the difficulties that OBD regulations may change frequently as they are driven by politics and differences over the nations.

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Overview

The DCM

- Ensures diagnostic data flow
- Manages the diagnostic states
  - Diagnostic sessions
  - Security states
- Checks that the diagnostic service request is supported
- Checks that the service may be executed in the current session and with the current security state

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Overview

- In the AUTOSAR architecture, the Diagnostic Communication Manager (DCM) is located in the Communication Services (Service Layer)
- In the Communication process, the DCM receives a diagnostic message from the PDU Router
- The DCM will check the diagnostic message
- Depending on the Diagnostic Service ID (SID), the corresponding calls in the Application Layer will be done
- Supported services
  - UDS
  - OBD
  - Flash Bootloader Interaction
    - Jump from App ←→ FBL, Flag Management

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VECTOR

Services for the Bootloader

- AUTOSAR 4 DCM may be used in Bootloader software since it supports already the needed services
- The AUTOSAR 4 DCM has APIs related to the Bootloader
  - `Dcm_SetProgConditions`
    - Store relevant information prior to
      - Jump to bootloader
      - Jump due to ECUReset request
  - `Dcm_GetProgConditions`
    - Called during `Dcm_Init`
    - allows to determine if a response ($50 or $51) has to be sent

Service Id	UDS Service	ASR 4.x
0x23	*ReadMemoryByAddress (RMBA)*	C
0x28	*CommunicationControl*	■
0x2C	*DynamicallyDefineDataIdentifier*	■
0x34	*Request Download*	C
0x35	*Request Upload*	C
0x36	*TransferData*	■
0x37	*RequestTransferExit*	■
0x3D	*WriteMemoryByAddress (WMBA)*	C
0x86	*ResponseOnEvent (RoE)*	■
0x87	*LinkControl*	Opt.

Block data transfer, also for bootloader

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Services for the Bootloader

Legend to the above table

- ☐ not supported (NRC Service Not Supported)
- ■ supported
- C callout (handwritten code, no service ID)
- Opt. optionally

Callouts for RMBA/WMBA

- Dcm_ReadMemory
- Dcm_WriteMemory

The AUTOSAR architecture doesn't provide the possibility to access the ECU memory using a physical address. This is implemented using a BlockId which identifies a memory block. Due to this fact, the DCM is not able to fully support the implementation of ISO14229-1 services which use a physical memory access.

As a solution, the DCM defines a callout to implement this kind of memory access. This callout implementation could simply be realized by defining a mapping between the BlockId and the physical memory address.

Further callouts of the DCM:

- Dcm_Confirmation – successful transmission or error during execution of a diagnostic service.
- Dcm_SetProgConditions – store Bootloader related information before jump into Bootloader
- Dcm_GetProgrConditions – read Bootloader information upon startup of ECU (Dcm_Init()) to determine if a RC 0x50 or 0x51 has to be sent
- Dcm_ProcessRequestTransferExit – DCM calls this if a download or upload shall be exited
- Dcm_ProcessRequestUpload – start upload
- Dcm_ProcessRequestDownload – start download

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Example for OBD related services

- The Service Range of OBD Diagnostics has a special Service Identifier (SID) range
- is prescribed by legislative regulations
  - Legislative diagnostics
- This OBD range does not have any intersection with the commonly known UDS services
  - Manufacturer diagnostics

SID	OBD Service
0x01	*Request Current Powertrain diagnostic Data*
0x02	*Request Power Train FreezeFrame Data*
0x03	*Request emission-related diagnostic trouble codes*
0x04	*Clear/reset emission-related diagnostic information*
0x06	*Request On-Board Monitoring Test-results for Specific Monitored Systems*
0x07	*Request Emission-Related Diagnostic Trouble Codes Detected during Current or Last Completed Driving Cycle*
0x08	*Request Control of On-Board System, Test or Component*
0x09	*Request Vehicle Information*
0x0A	*Request Emission-Related Diagnostic Trouble Codes with Permanent Status*

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DCM ComM Request

- DCM requests and releases communication directly through the ComM
  - During an ongoing communication service
    - DCM calls `ComM_DCM_ActiveDiagnostic` upon reception of the UDS request (`Dcm_TpRxIndication`)
    - DCM calls `ComM_DCM_InactiveDiagnostic` upon transmission of the UDS response (`Dcm_TpTxConfirmation`)
  - Upon transition into a non-default session
    - `ComM_DCM_ActiveDiagnostic`
  - Upon transition into the default session
    - `ComM_DCM_InactiveDiagnostic`
- Communication Control service is an exception; this is handled by BswM.
- Additionally DCM notifies the BSWM about its current Mode using the `BswM_DCM` API
- An ongoing diagnostics session keeps the communication channels requested
  - In default session, you will only have a COMM Request between sending the UDS Request and the UDS Response from the ECU
  - If you are in a higher ("non-default") session
    - it will be kept alive by the Tester Present Service
    - You should observe a permanent COMM Full Communication Request during the whole stay in this session

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Mode Management

- The DCM sends mode requests to the BswM based on the UDS requests it receives
- DCM acts as a Mode manager for
  - DcmDiagnosticSessionControl (service 0x10)
  - DcmEcuReset (part of service 0x11)
  - DcmModeRapidPowerShutDown (part of service 0x11)
  - DcmCommunicationControl_<symbolic name of ComMChannelId>. (service 0x28)
  - DcmControlDTCSetting (service 0x85)
  - DcmResponseOnEvent_<RoeEventID> (service 0x86)

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Example: DCM can request "Disable Normal Communication." During this mode BswM will turn off the corresponding I-PDU groups and NM PDUs.

Data use port

- Client / Server communication can be used for DIDs
  - Synchronous or asynchronous handling supported
  - DID Operations are implemented as Server Runnables in the application SWCs
- Sender / Receiver communication can also be used for DIDs
  - R-Port Interface for DcmDspDidRead
  - P-Port Interface for DcmDspDidWrite
- Scaling at Ports
  - The CompuMethod for the data type can optionally be derived from the DcmDspDiagnosisScaling

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Sender/Receiver and Scaling supported since MSR4 R17

DCM buffer

- The DCM buffer must be at least as big as the biggest DID for service 0x22 RDBI / 0x2E WDBI
- There is also a paged buffer support but this is dedicated for service 0x19 (Read DTCs) and its size is also determined by the biggest extended data record

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Agenda

- Diagnostics
- CANTP – CAN Transport Layer
- DCM – Diagnostic Communication Manager
- **DEM – Diagnostics Event Manager**
- FIM – Function Inhibition Manager
- Coming up next

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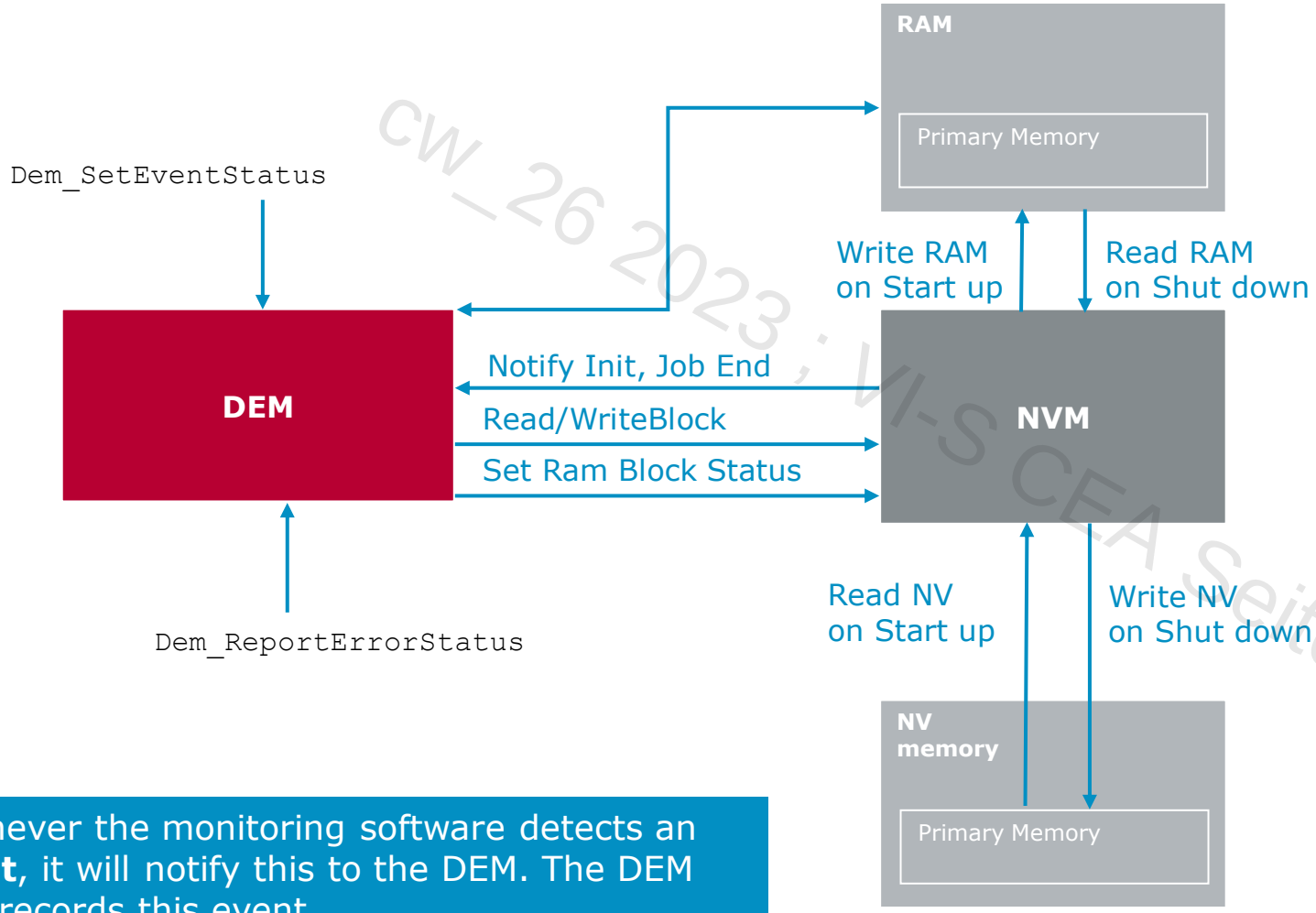
Module operation



Whenever the monitoring software detects an **event**, it will notify this to the DEM. The DEM then records this event.

SWCs normally signal fault Events to the DEM by invocation of the Dem_SetEventStatus() API. However not only SWCs can report Events to the DEM. The Basic Software Modules can also set "internal" Events by using the Dem_ReportErrorStatus() API.

The DEM uses various NvM Blocks to store its data:

AdminDataBlock: timestamps and operation cycle counter are saved every driving cycle

StatusDataBlock: Event data

DemPrimaryDataBlock0..n: The number of remembered DTCs (primary or secondary).

The DEM triggers NVM Block operations during runtime, but some blocks like StatusData and AdminData are only marked as valid and changed with NvM_SetRamBlockStatus API and actually written upon shutdown in NvM_WriteAll().

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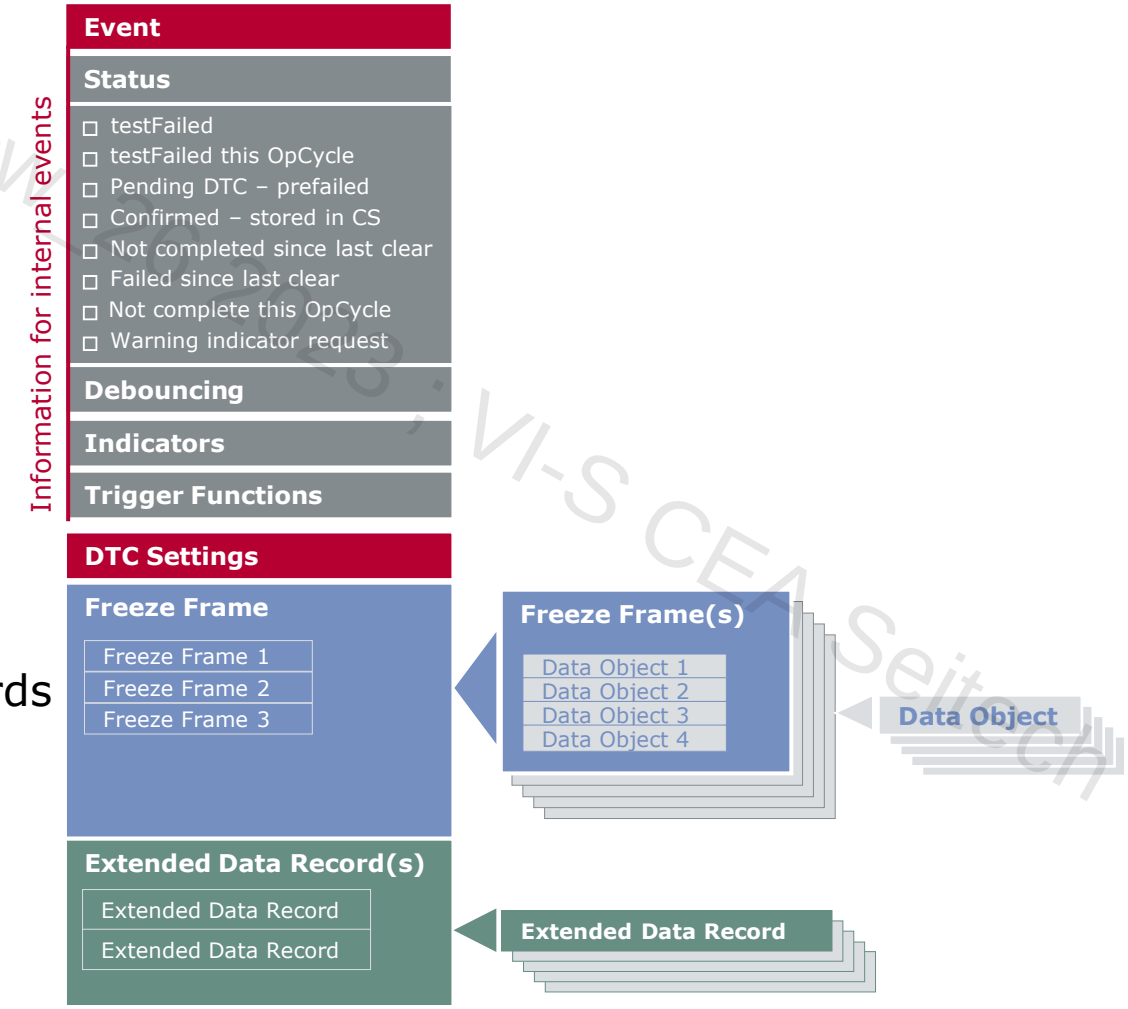
Events

- Events have a status
- "Learning" and "Forgetting" faults: (pre-)debouncing
- Indicator
- DTC information
- Additional information to specify an event
  - Freeze Frames
    - Freeze frames have data objects
  - Extended data records



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Events

Freeze Frame:

OEM-specific set of the vehicle/system operation conditions at a specific time. The term “Freeze Frame” is defined as a record of data (DIDs / PIDs). ISO 14229-1 refers to FreezeFrames as “SnapShotRecords”. There is the possibility to use pre-stored FreezeFrames. It means the data is already available and does not need to be acquired when the event status changes. This way time-critical Freeze Frame data can be saved together with the event.

Extended data records:

OEM-specific additional information assigned for a distinctive event which is not contained in the Freeze Frame and also described in ISO 14229-1. Extended Data Records are implemented by callbacks to the application. These callbacks are issued by the DEM at the point in time when the Diagnostic Monitor sets the error event in the DEM. The diagnostic configuration can provide 1..n extended data records for a DEM event. Depending on the point in time, different extended data records will be recorded together with the error event. In this way the first occurrence and the last occurrence can be considered.

Example for data not contained in Freeze Frames are: statistical information like **most recent** occurrence counter and **most recent** operation cycle / aging counter (they change over time). The same information contained in Freeze Frames would not change and is **“frozen”**.

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Event Storage

Each DTC (Diagnostic Trouble Code) is coupled with one byte providing status information. The bits represent the following meanings:

testFailed (bit0)
If the testFailed bit is set it means that the fault is active. The test for an error condition or wrong behavior has succeeded.

testFailedThisOperationCycle (bit1)
This bit is only set if the fault occurs in the current operation cycle. If the fault is active and this flag is not set it means that the fault occurred in a previous operation cycle.

pendingDTC – prefailed (bit2)
Indicates that the corresponding DTC will be set in the near future.

confirmedDTC – stored in Primary Memory (bit3)
As soon as the fault is stored in the Primary Memory, this bit is set. If the fault is cleared from the event storage the value of this bit will be changed too.

testNotCompletedSinceLastClear (bit4)
If there is no information about whether the DTC test has passed or failed since the last clear, this flag is active.

testFailedSinceLastClear (bit5)
This flag informs you that the DTC has set since the last clearing action.

notCompletedThisOperationCycle (bit6)
This bit corresponds with the NotCompletedSinceLastClear. It informs whether the complete information was set or reset in the current operation cycle.

warningIndicatorRequest (bit7)
Indicates whether the occurrence of this DTC should set a warning indicator.

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Event Storage

To provide the tester with more information about the point in time a DTC occurred, additional information is stored. This information is OEM-specific and is split into standardized and optional data.

Examples for OEM-specific standardized data:

Occurrence Flag
Distinguishes an entry as occurrence or fault

Original Odometer
This is the odometer value when the fault occurred for the first time.

Most Recent Odometer
This is the odometer value when the fault occurred the most recent time.

Frequency
This is the total number of times the DTC status has transitioned from “stored – not active” to “active.”

Operation Cycle Counter
This counter shows the amount of operation cycles (e.g., a new cycle starts after ignition off, ignition run, then wait 10 seconds). This information is used for a so-called self-healing process. If a DTC is stored but not active for a defined amount of operation cycles, it will be cleared from memory if it supports healing.

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Pre-debouncing

- The DEM can debounce Events reported to it via API

```
Dem_SetEventStatus(Dem_EventIdType EventId, uint8 EventStatus)
```

- Argument values of EventStatus for pre-debouncing:
  - `DEM_EVENT_STATUS_PREPASSED`
  - `DEM_EVENT_STATUS_PREFAILED`
- Algorithms for pre-debouncing
  1. Counter based
  2. Time based
  3. Debounce monitor internal[1]

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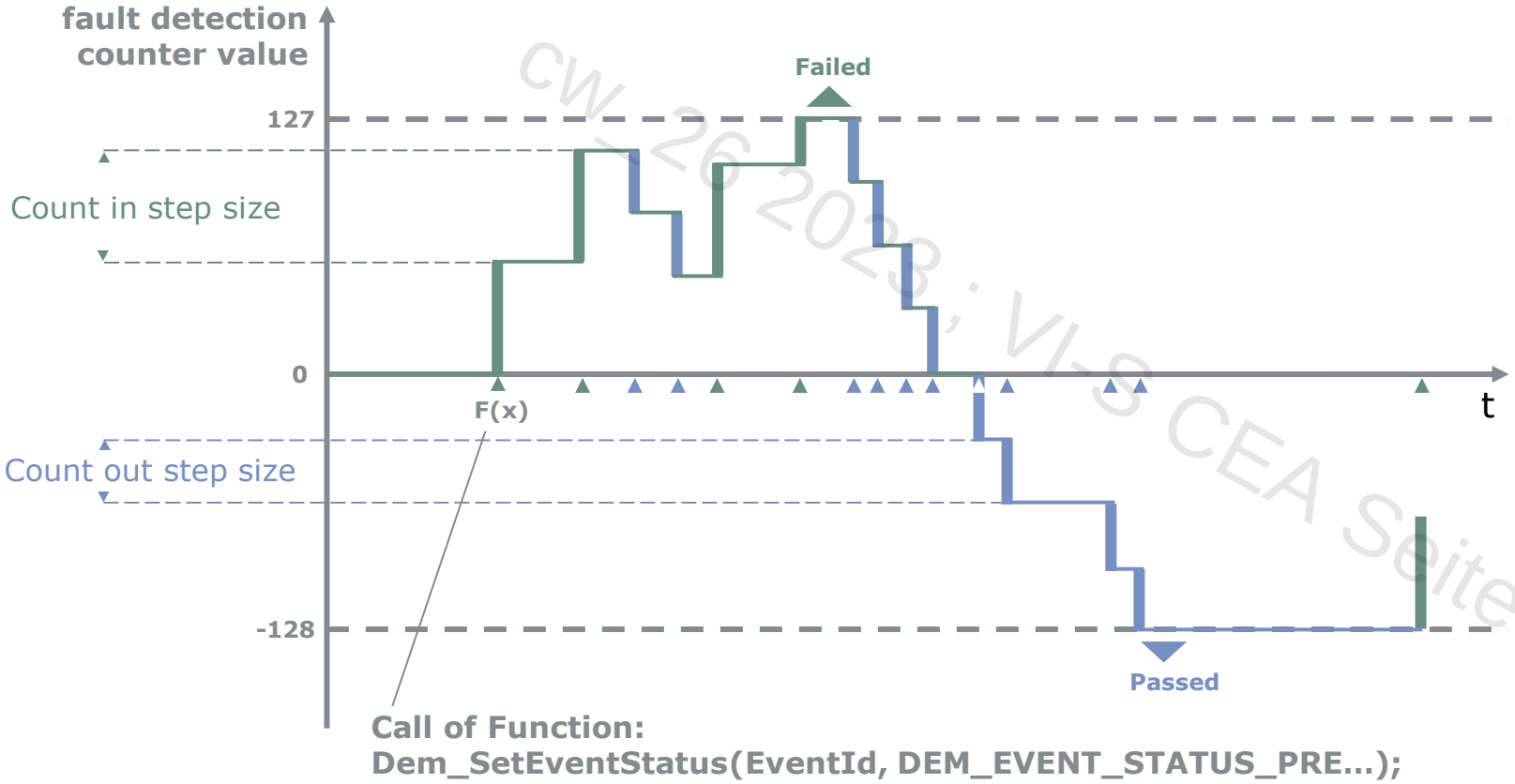
[1] Monitor internal debouncing is used for monitors that implement the debouncing themselves, and only report qualified results (PASSED/FAILED) to the DEM.

Pre-debouncing: Counter based

Events up – down counter



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Pre-debouncing: Counter based

Function Dem_SetEvent() can be called with the following parameter values

- `DEM_EVENT_STATUS_PASSED`: No debouncing, event is set to passed.
- `DEM_EVENT_STATUS_FAILED`: No debouncing, event is set to failed.
- `DEM_EVENT_STATUS_PREPASSED`: Debouncing relevant, configured debouncing done.
- `DEM_EVENT_STATUS_PREFAILED`: Debouncing relevant, configured debouncing done.

Current fault detection counter value can be requested via service port (Vector extension).

In the previous picture:

PREFAILED indicates the application calls

```
Dem_SetEventStatus(EventId, DEM_EVENT_STATUS_PREFAILED)
```

PREPASSED indicates the application calls

```
Dem_SetEventStatus(EventId, DEM_EVENT_STATUS_PREPASSED)
```

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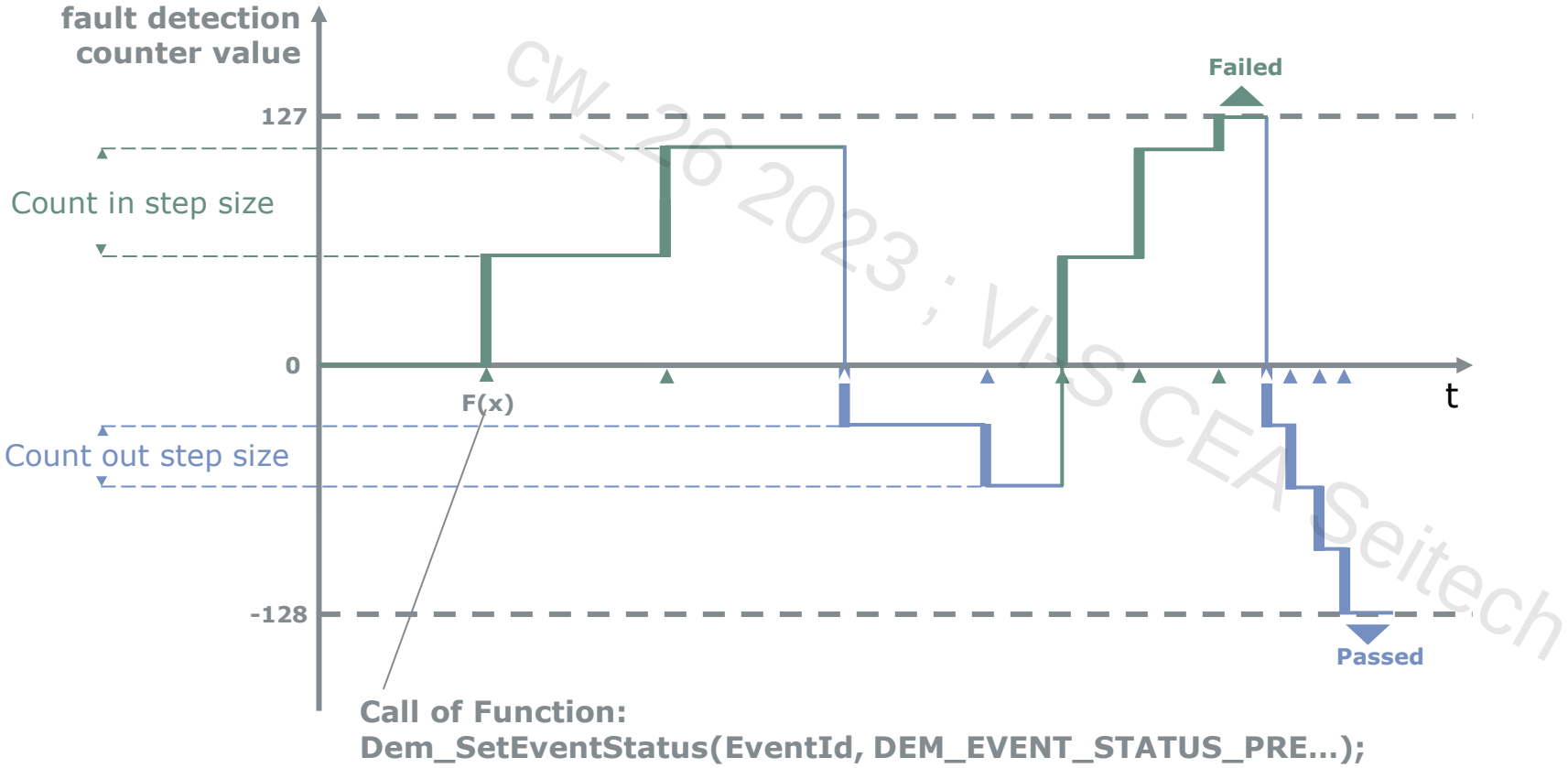
Pre-debouncing: Counter based

Events in a row counter



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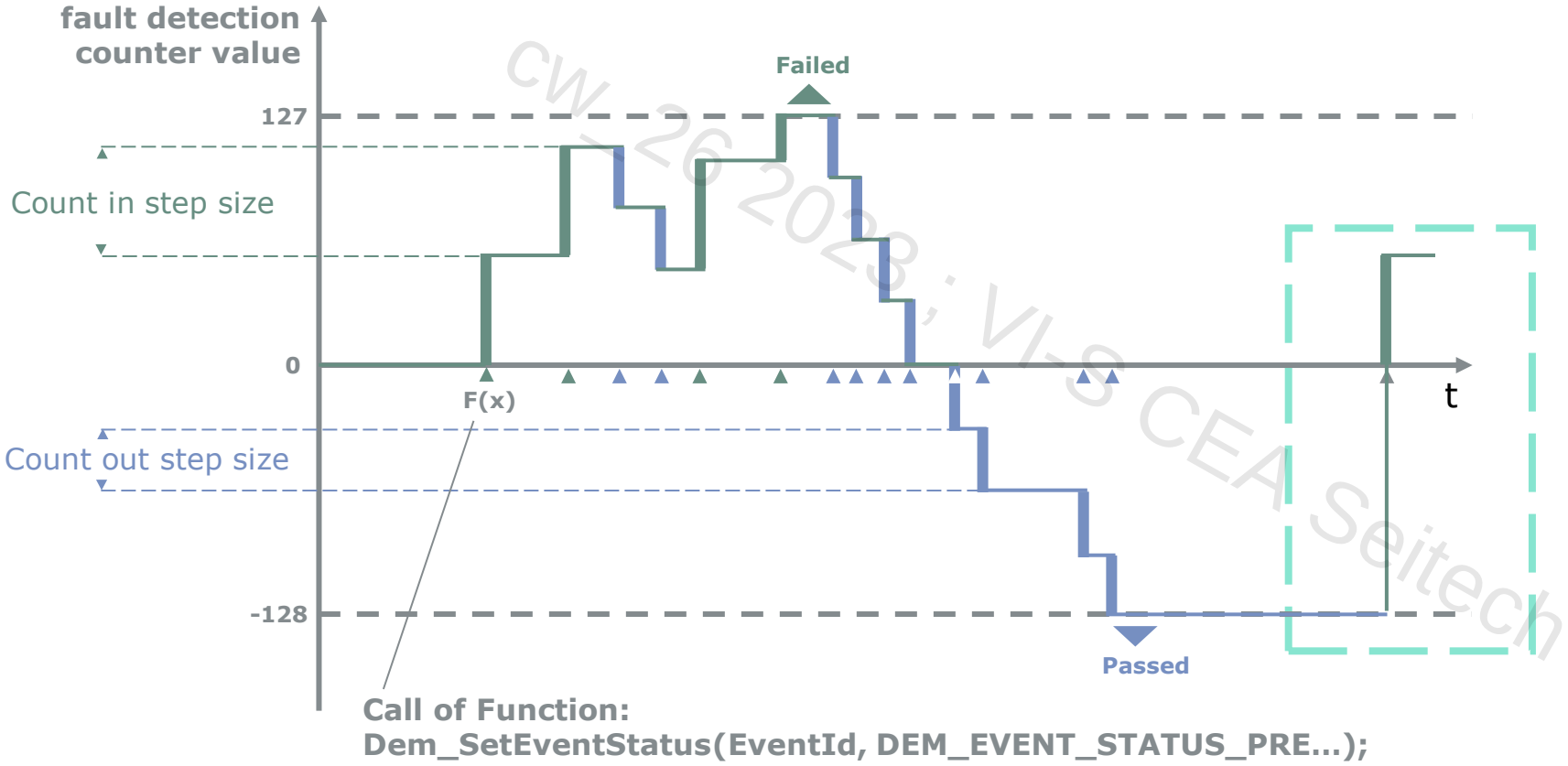
Jumps back to zero if new Dem_SetEvent argument is different from last call.

Pre-debouncing: Counter based

Count-in – Count-out/Jump-in



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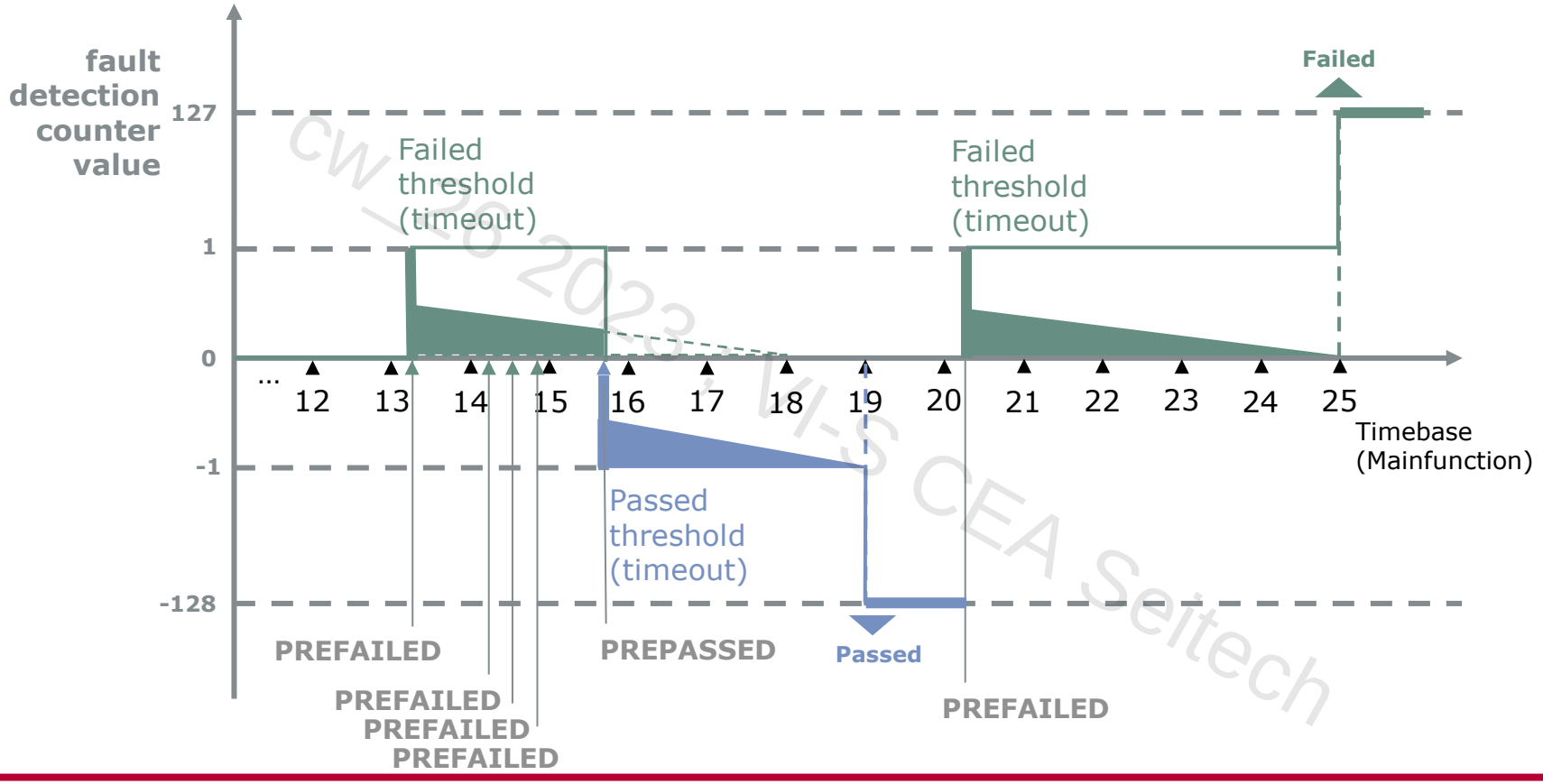
Pre-debouncing: Time based



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Failed threshold and passed threshold can be defined individually per DTC and do not necessarily have the same value.

In the above picture:

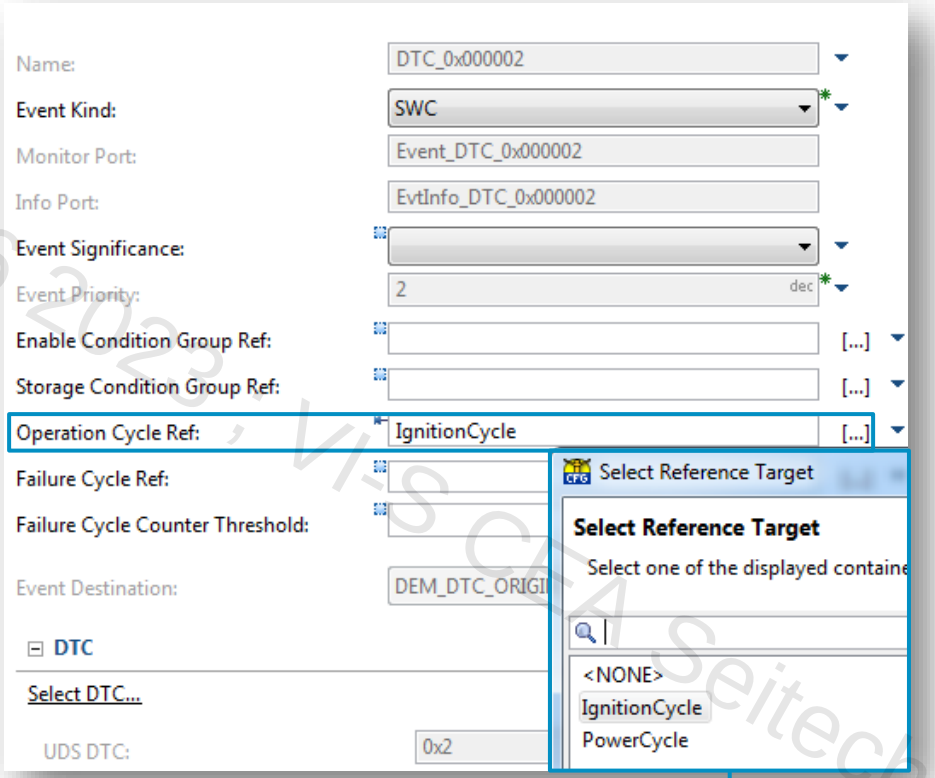
PREFAILED indicates the application calls `Dem_SetEventStatus(EventId, DEM_EVENT_STATUS_PREFAILED)`

PREPASSED indicates the application calls `Dem_SetEventStatus(EventId, DEM_EVENT_STATUS_PREPASSED)`

Operation Cycles – Events – Mechanisms

Examples of operation cycles are

- driving cycle
- engine warm up cycle
- ignition on/off cycle
- power up/power down cycle
- operation active/passive cycle
- accumulated operating time



Configurable operation cycles

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Internal and external events

- EventKind BSW events
  - Typically not accessible over UDS
  - The BSW reports them as Production Errors
  - Defined by AUTOSAR
- EventKind SWC events
  - Defined by the application
  - Typically accessible over UDS
- If you assign a DTC to an event, it can be queried over UDS
  - you can also equip an existing event (especially EventKind BSW) with a DTC by redefining it in the Diagnostic description file (CDD, DiagEx)

Dem Events	Event Kind	Monitor Port	Operation Cycle Ref
DTC_0x000002	SWC*	Event_DTC_0x000002	IgnitionCycle
AutoCreatedDemEvent_EcuMConfiguration_ECUM_E_CONFIGU	BSW		PowerCycle
AutoCreatedDemEvent_EcuMConfiguration_ECUM_E_RAM_CH	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_INTEGRITY_FAILED	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_LOSS_OF_REDUNDANCY	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_QUEUE_OVERFLOW	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_REQ_FAILED	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_VERIFY_FAILED	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_WRITE_PROTECTED	BSW		PowerCycle
AutoCreatedDemEvent_NVM_E_WRONG_BLOCK_ID	BSW		PowerCycle

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Internal and external events

Operation Cycle Kind of operation cycle for the event storage (e.g. power cycle, driving cycle, ...)

Healing Support Switch to allow healing/unlearning or not.

Healing Cycles Number of Operation Cycles necessary to heal/erase event.

Monitor Initialization Function Function which resets the error monitoring in the application

Event Destination

Bit 0: DEM_DTC_ORIGIN_PRIMARY_MEMORY

Bit 1: DEM_DTC_ORIGIN_SECONDARY_MEMORY

Bit 2: DEM_DTC_ORIGIN_MIRROR_MEMORY

Set the according Bit to '1' means event is assigned to the origin.

If an event is not assigned to an origin, then it is only handled internally and it is not visible externally.

Multiple origins are not supported.

Priority Priority of an event. This priority comes into play if the event memory is full. The priority decides if the new event is lost or if it overwrites a lower priority event in the memory.

DTC Number Diagnostic Trouble Code is a numeric code which consists of 3 bytes unique identifier for the error event. Together with this information, the status byte is also always saved. That means the minimum memory consumption of a DTC is 4 bytes.

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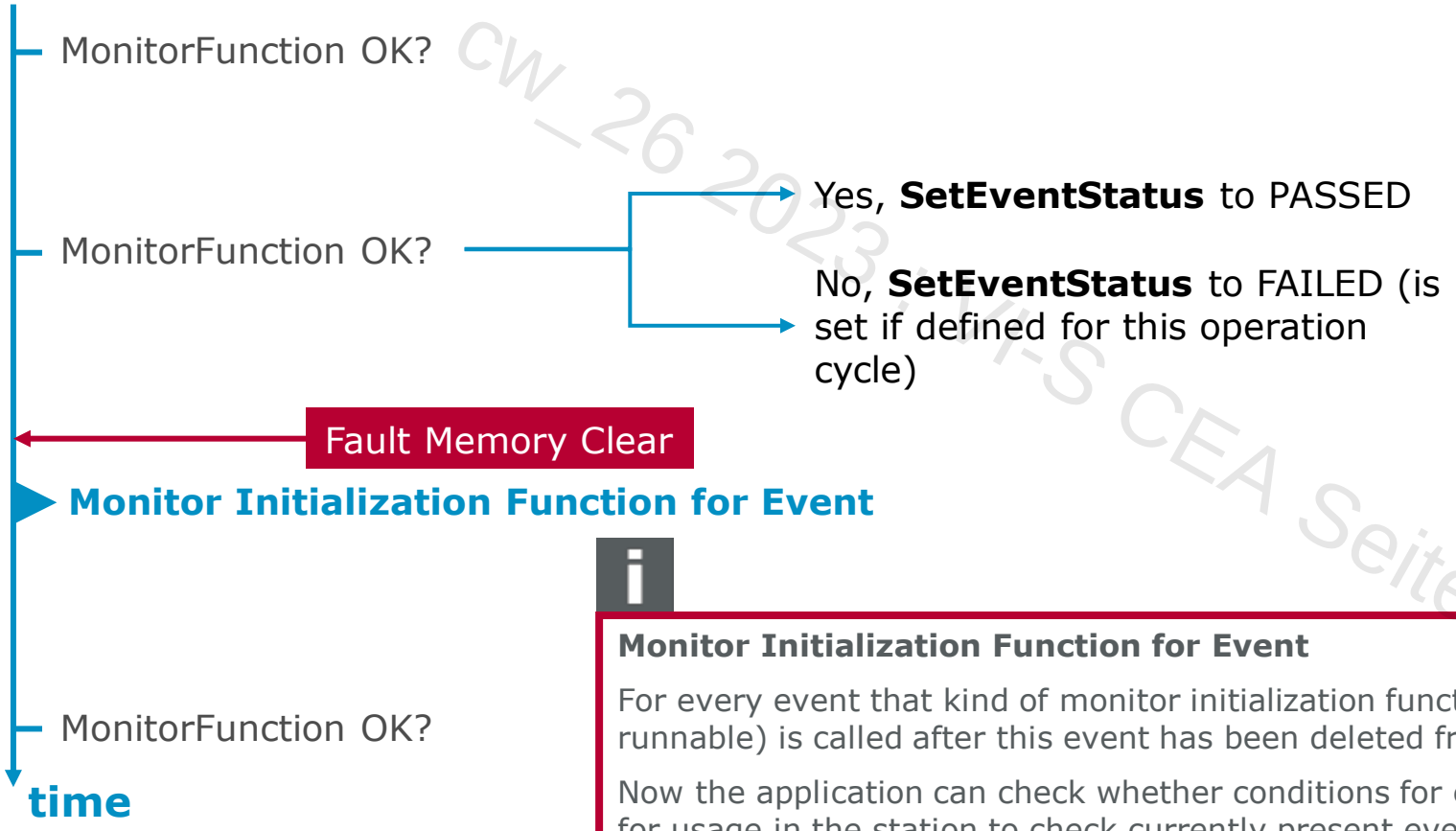
Diagnostic Monitor Initialization



After event is cleared a monitor initialization function is called.

The application can check whether conditions for this event are still there or not.

Monitor Initialization Function for Event

For every event that kind of monitor initialization function can be defined. This function (a runnable) is called after this event has been deleted from DTC memory.

Now the application can check whether conditions for entering this event are still present (e.g. for usage in the station to check currently present events).

Normally the monitor functions have large call cycles and may only be called seldom. With this initialization function, reaction time to a still-present event is very short.

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OBD functionality

- OBD functionality services $01 - $0A defined in SAE J1979 Rev May 2007
  - light duty OBD regulations OBDII, EOBD, Japan OBD, all others
- Only Legislative FreezeFrame (record 0) supported, records 1 and above are ignored
- The DEM calculates some PIDs internally (see next slide)
- The PID 0x0D Vehicle Speed is required for all OBD ECUs

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OBD functionality

Some details on the interaction between DEM and specific SWC might remain open, since they are dependent on the DEM and SWC implementation.

The following functionality is not defined:

- Malfunction Indicator Lamp (MIL)-activation (interface DEM to MIL handler, MIL-bulb check, readiness blinking, blinking in case of catalyst damaging misfire...)
- Misfire fault handling (common debouncing, filtering single / multiple misfire faults).

However the DEM does not prescribe implementation details on how OBD compliance can be achieved within the DEM module, e.g. concerning state handling. Furthermore, the DEM does not prescribe implementation details on the diagnostic algorithms of the SWC necessary to achieve OBD compliance (how to detect a fault, when to trigger increment of IUMPR-numerator...).

IUMPR = In-Use-Monitor Performance Ratio

Internally calculated PIDs

PID $01 – Monitor status since DTCs cleared (4 byte)

PID $02 – FreezeFrame DTC (2 byte)

PID $21 – km with MIL On (2 byte)

PID $30 – number of Warm Up cycles (WUC) since last fault clear (1 byte)

PID $31 – km since last fault clear (2 byte)

PID $41 – Monitor status this driving cycle (4 byte)

PID $4D – time with MIL On (2 byte)

PID $4E – time since last fault clear (2 byte)

PID $1C – OBD requirements to which vehicle or engine is certified. (1 byte)

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SRDataServices

- Sender Receiver Communication support
  - Port Interface SRDataServices_<SyncDataElement>
  - One Port Prototype per DID, PID, Extended Data Record Interface

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Agenda

- Diagnostics
- CANTP – CAN Transport Layer
- DCM – Diagnostic Communication Manager
- DEM – Diagnostics Event Manager
- ▶ **FIM – Function Inhibition Manager**
- Coming up next

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The FIM knows which functions are inhibited at the moment

- The application has to ask the FIM whether a certain function can be run or not.
  - FIM either stores the condition for each function identifier (FID) and is triggered by DEM on occurrence of an event
  - Or FIM requests the current events state from DEM and evaluates the condition for function identifier on request.

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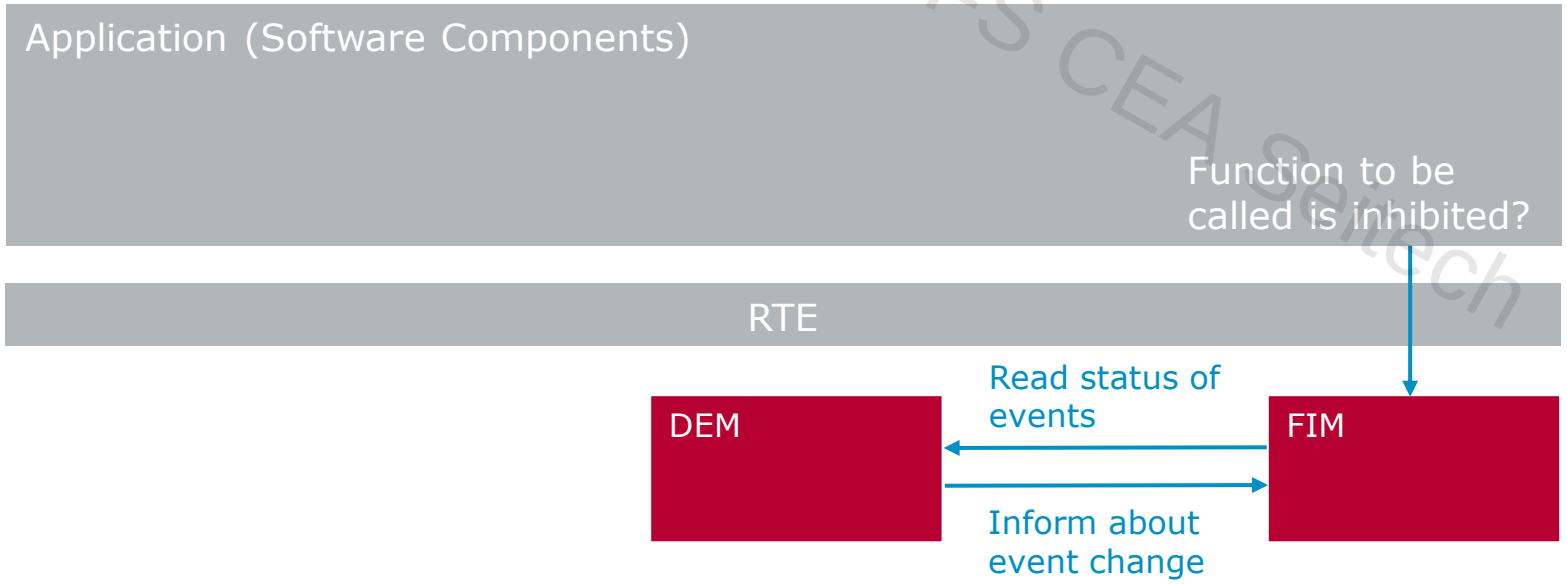

Diagnostics

Exercise: Diagnostics

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